MscFE 690 Capstone Project: Modue 6

Project Proposal– Student Group 17830

Authors: Ting Ting Han, Henry Mayambu, Jonathan Matura

ESG Investing and Stock Market Returns: Evidence from South African Mining Equities

Abstract

This study quantitatively assesses the materiality and risk adjusted performance of mining stocks listed on the JSE from January 2014 to June 2025 with respect to their Environment, Social and Governance (ESG) materiality and risk levels. Mining accounts for around one third of the market capitalisation of the JSE, around 8% of South African GDP, but is under heavy sustainability scrutiny in the context of the Mining Charter III, King IV Code, the carbon tax and the JSE Sustainability and Climate Disclosure Guidance (June 2022). The study fills three gaps in the literature, (i) the published evidence is mainly from developed markets, which is not applicable in emerging markets where loadshedding, commodity price volatility and regulatory changes are key factors; (ii) the published evidence is based on composite ESG scores, which mask pillar-level trade-offs; and (iii) the published evidence is look-back, because annual ESG scores are not aligned with monthly returns. The study uses two-way fixed-effects panel regression, Instrumental Variable GMM (IV-GMM) for endogeneity, Disaggregated Pillar-level Wald tests for materiality, equal-weighted tercile portfolio sorts with CAPM and Fama-French three-factor alpha estimation, and asymmetric GARCH-X modelling for systematic risk and downside risk. The 10 firms have about 110 independent firm-year ESG observations corresponding to 1,233 firm-month return observations, and the standard error is clustered by firm, giving the score at time of the return an explainable role in the return without look-ahead bias; the ESG observations are clustered by firm, while the return observations are clustered by month. There are 11 mining companies listed on the JSE that are part of the final tradeable universe. Finally, the study introduces, in a localized and open source manner, a framework of factors for the JSE mining sector which can be used by capital allocators as per Regulation 28 of the Pension Funds Act as a rigorous and a risk adjusted tool to support sustainable investment decisions.

Keywords: *ESG investing, stock market returns, JSE mining equities, panel econometrics, GARCH volatility, sustainable finance, emerging markets, South Africa.*

1. Introduction

1.1 Context and purpose

The world of investment has gone from a checklist of ethics to an ethical duty of the investor. Mining is an integral part of South Africa's economy, accounting for about one-third of the South Africa Stock Exchange (SX-SE) market capitalisation, and contributing about 8% to South Africa's Gross Domestic Product (GDP) (Matos), but it also has a heavy burden of sustainability problems, including strict carbon and water regulations, strict labour laws, and socio-economic requirements, such as Broad Based Black Economic Empowerment (B-BBEE) and Mining Charter III compliance (Matos). As investors in the JSE have become more vocal about their desire for transparency with regards to ESG compliance, it is empirically still unclear whether the ESG performance generates or destroys value in this labour and carbon intensive industry.

This enquiry is framed by two perspectives of theory. Friedman's (1970) "Shareholder Primacy" model sees discretionary ESG spending as an agency cost which hurts short-term liquidity and stock prices (Edmans). The modern risk mitigation models of Freeman (1984) describe material ESG risk management as an investment in reputational capital that lowers the cost of capital in the longer term and safeguards against reputational harm and regulatory penalties. Good ESG practice could reduce the cost of capital and enhance returns in capital-intensive and politically-exposed sectors by reducing risk.

1.2 Research Problem

Although sustainable investing is becoming more and more popular, three gaps in the literature remain. First, the evidence available is mainly from developed Western markets, results from stable economies may not be applicable to economies in transition, emerging markets, or markets with a high degree of load shedding, volatile commodity prices and changing regulations (Kräussl et al.). Second, there is a South African mining paradox – investments in ESG (renewable energy as a solution to loadshedding, safety upgrades, community projects mandated under the Mineral and Petroleum Resources Development Act (MPRDA) and carbon tax are expensive – but can be a critical long-term hedge against fines and reputational damage. It is unclear what the reaction of the market will be, if any. Finally, the majority of studies lump ESG scores into a "black box. Recent research shows that returns are influenced differently by each pillar (Giese et al.): environmental and governance factors have an effect in the short-term, social factors have an effect in the long-term. Local level consensus on the concept of materiality for companies listed on the JSE in the mining sector is lacking (Berg et al.).

Fourth, there is a lag time between the annual Refinitiv scores and the monthly returns they are designed to explain. The scores are prepared about six months after fiscal year-end, which means that by reporting the annual score at the same time as the monthly returns for the same calendar year, look-ahead bias is

created. This study overcomes this by giving each annual score the credit it deserves after the month it was published, and assigning the monthly return to the latest annual score published, while clustering SEs at the firm level by using merge_asof.

A central research question is whether and to what extent ESG performance at the pillar level impacts the risk-adjusted returns of JSE-listed miners, without a local guide.

1.3 Research Questions

- RQ1: Is there a difference in the impact of the individual E, S, and G pillars on stock returns?
- RQ2: Can one conclude that ESG performance influences systematic risk and downside volatility of South African mining portfolios?

1.4 Hypothesis

Table 1: Hypothesis

Hypothesis	Statement	Null	Alternative	Test
H1	Composite ESG affects monthly excess returns	$\beta_ESG = 0$	$\beta_ESG \neq 0$	Two-way FE PanelOLS, clustered SEs
H2	Pillars have differential effects	$\beta_E = \beta_S = \beta_G$	At least one differs	Disaggregated panel + Wald test
H3	High-ESG portfolios earn positive risk-adjusted returns	$\alpha = 0$	$\alpha \neq 0$	CAPM and FF3 alpha estimation
H4	ESG reduces systematic and downside volatility	$\theta_ESG = 0$	$\theta_ESG < 0$	Asymmetric GARCH-X variance equation

1.5 Significance

The study can be helpful to fine-tune emerging market resource portfolios for asset managers, as well as mining companies thinking about sustainability investments. It pulls inferences from 11 exchange-listed mining companies (10 companies included in the estimation panel) and results in some 110 independent firm-year ESG observations, which can be generalized for the sustainable finance literature in Africa, and inferences can be drawn at the level of the pillars. The study contributes to the literature by explicitly addressing the issue of the timing alignment of the annual ESG scores and monthly returns and by taking into account the clustered standard errors at the firm level. It also provides a local and open source factor framework adapted specifically for the JSE-listed equities market, which is missing from the market for the institutional investors subject to Regulation 28 of the Pension Funds Act.

2. Theoretical Framework

2.1 Shareholder versus Stakeholder Theory

In the theory of corporate sustainability, there are conflicting views on the main goal of the company. Based on neoclassical economics, a company's primary obligation is maximising shareholders' profits. From this view, the economic cost of allocating funds to voluntary ESG initiatives is the increase in costs for compliance, administration and competitive disadvantage, which lowers short term liquidity and stock prices (Friedman; Pedersen et al.; Edmans).

The stakeholder theory, however, holds that creating value over the long-term is a matter of balancing the interests of multiple stakeholder groups – employees, the community, regulators, and environmental systems (Freeman). It also builds on that with modern risk-mitigation models arguing that having a strong ESG integration is a protective barrier against extreme financial risks, regulatory penalties and reputational damage (De Klerk and De Villiers). Better ESG practices reduce the cost of capital and enhance returns in capital intensive and politically sensitive industries via risk-reduction mechanisms.

2.2 Pillar-Specific Materiality

The ESG scores can be confusing because they are combined into an overall score. For mining equities, the dynamics that need to be considered under the E pillar are carbon transition risk exposure, regulatory compliance, and ecological damage liability (De Klerk and De Villiers). Environmentally intensive industries are typically characterized with negative environmental factor scores (environmental capital expenditure is high, future carbon taxes are uncertain, asset obsolescence is possible): stocks in such industries are more volatile, although effective environmental policies serve as a safeguard against the effect of future carbon taxes and asset obsolescence (Avramov et al.).

Social (S) and Governance (G) pillar dynamics are key to obtaining and maintaining a Social License to Operate (SLO) in emerging markets. Social issues may result from unsafe working conditions, unequal pay, and compelled labor that can trigger violence or closure of the site (Bauer & Hann, 2010). Governance mechanisms that limit information asymmetry, boost institutional investor confidence, and improve minority investor protections lead to higher valuations (Tobin's Q) for developing-economy companies. Khan et al. (2016) expanded this by identifying material and immaterial ESG factors, based on the SASB framework, and indicated that value creation is associated with material ESG factors, rather than with disclosures.

2.3 Empirical Evidence and Cross-Market Findings

The evidence on the empirical outcomes of the integration of ESG is mixed. Dimson et al. (2015) find a wide range of evidence, but it is clear that, in developed markets, active ESG engagement can significantly lower downside risk. For extractive industries, especially, financial impacts have been found to be contrary: Environmental liabilities raise credit risk and cost of debt financing, as pointed out by Bauer and

Hann (2010). De Klerk and De Villiers and Johnson et al., indicated that the codes of good governance that were classified under King III/IV had positive relationship with improved disclosure, but disclosure was not always associated with market valuation premiums in South Africa. The final answer is therefore that as long as ESG practices lower risk, they do not directly help in return enhancement. One limitation is that composite scores are used for ESG instead of disclosing what is happening at the level of the pillars.

2.4 Methodological Development

Recent developments move away from simple linear regression models to dynamic, non-linear, and regime-specific models. Pedersen et al., proposed the ESG efficient frontier, which shows the impact of the ESG restrictions on the efficient frontier. Avramov et al., introduced uncertainty of ESG ratings into the equilibrium pricing model. Bolton and Kacperczyk introduced carbon beta and Just Transition risk, based on the notion of a persistent global carbon premium. After hitting these quantitative thresholds, this project discovers ESG alpha with the help of Refinitiv Asset 4 pillar scores and Fama French 5 factor and momentum models.

2.5 Mixed Empirical Evidence and Methodological Challenges

There are inconsistencies between studies in both countries and methods. Empirical divergence and rating disagreement suggest that longer time horizons might better detect a relationship between stock prices and sustainability ratings, although no statistically significant correlation is observed for shorter time periods, and conflicting results exist on whether positive or negative relationships exist (Avramov et al.; Blitson; Bolton and Kacperczyk). The differences are primarily due to the use of different methodologies for rating (Bloomberg, MSCI, Sustainalytics), sampling time periods, and specification errors in the models. Prior studies have been criticised for their lack of attention to the econometric rigour, endogeneity and unobservables (firm-level heterogeneity and reverse causality). In recent years, panel data estimators like the Fixed Effects (FE) model, robsust standard error clustering, and system Generalized Method of Moments (GMM) have become more popular to control for firm-specific factors and thus to obtain more precise causal effects (Wintoki et al.).

2.6 Where this Study Fits

Three-Part Empirical Gap:

- No Sector Focus: Cross sectional JSE studies do not consider dynamics in the extractive sectors.
- Aggregation Bias: Composite ESG scores ignore conflicting trade-offs between the E, S and G components.
- Uncaptured Regulatory Realities: Current studies do not take the developments up to 2025 into

account, such as the Mining Charter III, the King IV and Carbon Border Adjustment Mechanisms (CBAM).

Dynamic panel estimators (two-way FE and IV-GMM) control for dynamic endogeneity, and augmented Fama-French 5 factor + momentum framework (Carhart 6 factor model) is integrated with a disaggregated Refinitiv Asset4 pillar model.

2.7 Competitor Analysis

Table 2: Competitor Analysis

Analysis Dimension	De Klerk et al.	Johnson et al.	Global Benchmark (Bauer and Hann; Bolton and Kacperczyk)	Our Study (Proposed)
Focus & Scope	CSR disclosure metrics, JSE Top 100	Integrated reporting quality (King III/IV), wide SA stocks	Global/non-EM markets, environmental liabilities, carbon emissions	Performance-based ESG integration in JSE-listed mining stocks
Data & Timeline	Voluntary disclosure scores (pre-2018)	Post-2015 reporting quality	Global/US-centric datasets	Post-reform window (2014–2025): King IV, Mining Charter III, carbon transition
Primary Method	Cross-sectional correlations, simple OLS	Panel regressions of Tobin's Q on disclosure quality	Traditional asset pricing (Fama-French) or credit spread regressions	Augmented Fama-French multi-factor + dynamic panel estimators (FE/IV-GMM)
Key Limitations	No risk-adjustment controls	Mixes disclosure with real ESG performance	Ignores local socio-economic dynamics	Strict treatment of sector-specific data and dynamic endogeneity
Our Value Add	Baseline comparison	Pillar disaggregation	Localization & asset pricing	Investable insights: mining-specific risk-adjusted framework

2.8 Business Case

Commercial and practical viability of this model is being supported by South African asset managers and

institutional investors in this country who are under tight regulatory constraints, specifically Regulation 28 of the Pension Funds Act. Although ESG factors are mandated to be incorporated into capital allocation decisions at the local level, there are no off-the-shelf global ESG factor models that can precisely value region-specific operational and regulatory risk in emerging markets. The project aims to meet a major market gap by delivering a factor framework that is open-source, localized and specific to JSE-listed mining stocks – a tool that institutional investors can use to meet compliance obligations and optimize portfolios while operating in a risk-adapted manner.

2.8.1 Strategic Market Positioning

- Aggregate vs Disaggregation: This paper will examine each pillar individually, with its own set of environmental, social and governance risks/penalties.
- Methodological Superiority: Advanced panel econometric techniques (accounting for endogeneity and unobservables of firms) yield data relevant for the analysis of the JSE-listed mining industry that are reliable.
- Bridging Theory and Practice: Links ESG disclosures to financial returns via Tobin's Q approach and fluctuation in stock returns analysis, giving informative information for efficient asset allocation in risky developing economies.

3. Methodology

3.1 Research Design

The ESG scores are sourced from the Refinitiv extract supplied for this study, namely 14 mining ticker symbols that are listed on the JSE, monthly, January 2014 to June 2025. The closing prices are obtained from Yahoo Finance for data that is adjusted on a monthly basis. The monthly closing prices are retrieved from Yahoo Finance, as are the closing prices of the JSE market index and the ZAR/USD exchange rate. Risk-free rates are sourced from the series IR3TIB01ZAM156N from the Federal Reserve Economic Data (FRED), which is the 3-month South African interbank rate, and are on a monthly frequency. The ESG and return series are then joined on (Ticker, Date) with the 6-month lag in ESG publication described below.

3.2 Data and sample

The Refinitiv ESG extract that was provided for this study includes 14 tickers, but only 8 are present in the design list (AGL.,.JO, IMP.JO, SSW.JO, HAR.JO, GFI.JO, EXX.JO, PAN.JO, and NPH.JO, listed as NHM.JO in the extract). The 6 ESG companies are on the design list (ANG.).JO, AMS.JO, CML.JO, MCG.JO, RBplatinum, SLM.JO). Seven design firms (S32.JO, ARI.JO, THA.JO, MRF.JO, DRD.JO, ASR.JO, KIO.The extract does not include details of JO). Three ESG firms are eliminated, after the filters of an ESG score in the extract from the Refinitiv and the price series from Yahoo Finance. Dedicated Laptop (no Yahoo data), MCG.JO (same corporate lineage as CML.Among the other delisted companies are JO (merged into Impala), RBplatinum (merged into Impala in 2022) and RBPLUS (merged into RBplatinum in 2022). So the final universe of tradable firms is 11 firms: the 8 firms designed matched firms plus ANG.JO, CML.JO and SLM.JO. In the two-way FE estimation, SSW. However, the panel is shortened in JO because Yahoo Finance only contains 64 month observations for it (from February 2020), while the other 10 firms have 1,233 firm-month observations.

Table 3: Table 3: Sample Reconciliation — 15 Design Firms to 11 Tradeable Firms

Category	Count	Firms
Design proposal	15	AGL, IMP, S32, SSW, HAR, GFI, EXX, ARI, THA, MRF, PAN, DRD, NPH, ASR, KIO
ESG file (all tickers)	14	AGL, AMS, ANG, CML, EXX, GFI, HAR, IMP, MCG, NHM, PAN, RBplatinum, SLM, SSW
ESG ∩ design	8	AGL, EXX, GFI, HAR, IMP, NHM (=NPH), PAN, SSW
ESG − design (additions)	6	AMS, ANG, CML, MCG, RBplatinum, SLM
Design − ESG (missing)	7	S32, ARI, THA, MRF, DRD, ASR, KIO
Tradeable on Yahoo (final)	11	AGL, ANG, CML, EXX, GFI, HAR, IMP, NPH, PAN, SLM, SSW
Not tradeable on Yahoo	3	AMS, MCG, RBplatinum

3.2.1 ESG Data Alignment and Timing

- Refinitiv ESG scores are calculated annually and published on a regular basis about six months after the fiscal year end. To avoid the look-ahead bias, the score is assigned to the panel from the month of its publication, so that a score from December 2020 is not used to explain returns for June 2020 until it is published a year later in December 2021. Companies with June and September fiscal-year-ends follow the same pattern (December and March).

- Specifically, the regressor for each monthly return observation in the panel is the latest annual ESG score published (merge_asof backward). The standard errors are localised around the firm level yielding 10 effective clusters. Each annual score is applied to 12 consecutive monthly returns, giving us about 110 firm-year scores (10 firms x 11 fiscal years), compared to 1,233 firm-month return observations. Refinitiv states that scores in the last five fiscal years are not definitive and subject to possible revision back in time; this is a source of look-ahead bias in the subsample of scores from FY2020 to FY2025, which is reported.

- Panel Structure: Unbalanced Panel. The raw ESG file has 1,850 firm-month observations for 14 tickers. The ESG panel has about 110 independent firm year observations from 10 firms, located on 1,233 firm month returns after collapsing to fiscal-year-end values and applying the six-month publication lag. The 82-month period of excess over the theoretical full grid (14 x 138 = 1,932) is indicative of listing and delisting events that are documented in §6.5.

3.2.2 Corporate Actions Adjustment

- The 2025 de-merger of Anglo American, taking its platinum business separate and combining the shares.

- The de-merger of South32 in 2015 from BHP Billiton.

- Price series unadjusted for price changes resulting from share consolidation and unbundling.

- CML.JO (Coal of Africa) and MCG.JO (MC Mining) are of the same corporate lineage with the exception of CML.JO is left in the final universe, and MCG.To avoid double counting, JO is excluded.

3.3 Model specifications

3.3.1 Model1: Two-Way Fixed Effects Panel Regression (H_1: ESG –Return)

$$R_{i,t} = \alpha 0 + \delta R_{i,t-1} + \beta 1 ESG_{i,t} + \gamma Controls_{i,t} + \mu_i + \lambda_t + \varepsilon_{i,t}$$

Where excess return for firm i at month t is R_{i,t} with t being the time observable, the annual ESG score for firm i (with a 6-month publication lag) is ESG_{i,t} and the entity (firm) fixed effects are μ_i, while the time fixed effects are λ_t and the error term is ε_{i,t}. Note on the market factor: The market return is not considered as a separate regressor since it is the same across firms in a given month and is thus completely captured by the time fixed effects λ_t. The market factor is instead added in the CAPM alpha regressions in model 4, with no time fixed effects. It would cause an absorbing-effect error if you added it to Model 1. Estimation Methodology: PanelOLS (EntityEffects and TimeEffects), clustered SE (at firm level). N = 1,233, 10 firms, 137 time periods. The joint significance of entity and time effects is confirmed by Poolability F-test (F = 15.04, p < 0.001).

3.3.2 Model 2: Instrumental Variable GMM (Endogeneity Corrections)

$$R_{i,t} = \alpha 0 + \beta 1 ESG_{i,t} + \gamma Controls_{i,t} + \varepsilon_{i,t}$$

Data for instrument: ESG (Previous FY) (12-month data lag (not monthly), not rolling)

ESG_lag1_{i,t} = ESG_{i,t−12}

Estimator: IVGMM (robust standard errors). Validity: If the model is validly determined (1 instrument, 1 endogenous regressor), then the Hansen J-test is not applicable, and the validity of the instrument is based on the exclusion restriction.

3.3.3 Model 3: Asymmetric GARCH-X (H4: ESG-Volatility)

In the arch library, exogenous regressors are not directly used in the variance equation of the GARCH models, so the estimation of Model 3 follows the two-step GARCH-X procedure of Bollerslev (1986).

- Step 1: Estimate a skewt-GARCH(1,1) with AR(1) mean per firm to get the conditional variance series $\sigma_2 t$.
- Step 2: Regress log(σ^2_t) on the lagged annual ESG score using the HAC standard errors (maxlags = 4).
- The coefficient of ESG in the second stage regression gives the variance equation's parameter, θ_ESG. For a negative, significant θ_ESG, ESG reduces volatility. A one sample t-test is used to test the cross-firm mean θ.

3.3.4 Model 4: Portfolio Sorts and Factor Alphas (H3: Risk-Adjusted Performance)

Tercile Portfolio Construction:

- Classify companies according to their composite ESG scores, monthly.
- Form equal-weighted tercile portfolios (High, Medium, Low ESG)
- Rebalance monthly
- Apha Estimation
- Use the long-short portfolio (High ESG – Low ESG) for alpha estimation.

CAPM:

$$R_{p,t} - R_{f,t} = \alpha p + \beta_p(R_{m,t} - R_{f,t}) + \varepsilon_{p,t}$$

Fama-French Three-Factor:

$$R_{p,t} - R_{f,t} = \alpha p + \beta_p(R_{m,t} - R_{f,t}) + s_p SMB_t + h_p HML_t + \varepsilon_{p,t}$$

The alphas reported are the three-factor alphas (Fama-French) which are not reported because the SMB and HML factors were not available for the JSE on the selected market-data provider and were set to zero as placeholders; the FF3 estimates are therefore the same as the CAPM estimates by construction. However, it is pointed out in §6.5 that this limitation is not a problem in the case of a small energy gap.

Validity: Alpha significance (t-test), Sharpe ratio, maximum drawdown.

3.3.5 Model 4b: Pillar-Level Regression and Wald Test (H2)

Two way FE panel regression with E, S, G as separate regressors and Wald test of the joint null hypothesis β_E = β_S = β_G based upon two pairwise linear constraints.

3.4 Data Analysis Procedure

Table 4: Data Analysis Procedure

Step	Procedure	Software	Output
1	Data acquisition (Yahoo Finance prices, Refinitiv ESG)	Python 3.11	Raw datasets
2	ESG alignment with 6-month publication Lag	pandas merge_asof	Aligned panel
3	Excess return and Sharpe ratio calculation	Pandas	Dependent variables
4	Descriptive statistics and scatter plots	Matplotlib	Illustrative figures

5	Model 1: Two-way FE panel regression	linearmodels.PanelOLS	β_ESG, R^2, F-test
6	Model 2: IV-GMM with lagged ESG Instrument	linearmodels.IVGMM	β_ESG, AR(2), Hansen J
7	Model 3: Asymmetric GARCH-X per firm	Arch	θ_ESG, AIC/BIC
8	Model 4: Tercile portfolio sorts and alpha Estimation	pandas, statsmodels	α, Sharpe, max drawdown
9	Pillar-level Wald test (H2)	Linearmodels	Wald statistic, p-value
10	Visualization and reporting	matplotlib, seaborn	Figures and tables

There are a number of structural limits to the data and methodology, including annual frequency of ESG scores and the lack of some risk factors. These main constraints, their potential impact on the analysis and methodological mitigations applied to reduce the impact, are summarised in Table 5.

Table 5: Constraints and Mitigation

4 **Constraint**	**Impact**	**Mitigation**
Unbalanced panel (firm entries/exits)	Reduced power in some periods	Report per-firm observation counts; robustness checks
Annual ESG vs monthly returns	Look-ahead bias, inflated N	6-month publication lag; firm-clustered SEs; report 165 independent ESG obs
Refinitiv retroactive revisions (FY2020–FY2025)	Look-ahead bias in recent subsample	Sensitivity analysis; flag as limitation
Small number of independent ESG observations (110)	Limited power for pillar tests	Wald test with adjusted degrees of freedom; report firm-month N = 1,233
Corporate actions (de-mergers, consolidations)	Price discontinuities	Restate price series; adjust for unbundling
Commodity super-cycle 2021–2022	Confounding volatility	Commodity-specific controls; time fixed effects
Loadshedding events	Idiosyncratic volatility	Time fixed effects; robustness with event dummies
Survivor bias from delisted firms (ASR, RBplatinum)	External validity	Documented exclusion; not a sampling choice
SMB/HML factors unavailable on JSE provider	FF3 identical to CAPM	Documented limitation (§6.5); FF3 reported for completeness only

3.5 Source Code Development

To help facilitate teamwork, version control and sharing between the team and instructor, the project code is hosted in a private Git repository on GitHub, following the guidelines outlined in the MScFE 690 Source Code Development guidelines. At the highest level, it includes a README.md file that provides a

description of the project, how to set up, and how it is executed; a requirements.txt file that lists down all the python packages needed so that the instructor can install everything once with the command pip install -r requirements.txt; a CONTRIBUTORS.md file that documents the roles of contributors and their responsibilities; a config.yaml file that specifies the ticker symbol, date ranges, fiscal year-end months, and model settings in a centralised manner; and a .gitignore file that excludes data files, outputs and credentials from version control.

The data is stored in two folders: data/raw includes the original Refinitiv ESG scores and JSE price data and data/processed includes the cleaned and aligned panel data for the analysis. All core Python code is in the src folder, broken into small modules: data_loader.py is for loading and acquiring the data, esg_alignment.py is for aligning by the six-month publication lag using merge_asof, returns.py is for the calculation of excess returns and Sharpe ratios, panel_models.py is for the two-way fixed effects model and IV-GMM estimation, garch_models.py is for the asymmetric GARCH-X estimation, portfolio_sorts.py is for the construction of tercile portfolios and estimation of alphas, and visualization.py is for the creation of figures and tables.

The notebooks folder includes four jupyter notebooks for a step-by-step guide to the analysis: data exploration, panel regression, GARCH analysis, and portfolio backtest. Results are stored in the outputs folder, the figures (as 300 dpi PNGs) in one subfolder, and the regression and summary tables in another. The last one is the tests folder which includes unit tests for the alignment part: test_alignment.py, and the models part: test_models.py, to ensure the reproducibility and correctness of the results.

The following structure follows the Source Code Development Guidelines: a private Git repository, full access for team members, read-only access for instructor, a requirements.txt file for installation of all dependencies, documentation (README, config, contributors). Full access link that can be shared with the instructor for review via HTTPS.

4 Results

4.1 Descriptive Statistics

Table 6: Descriptive Statistics of Returns, ESG Scores, and Controls (N = 1,233)

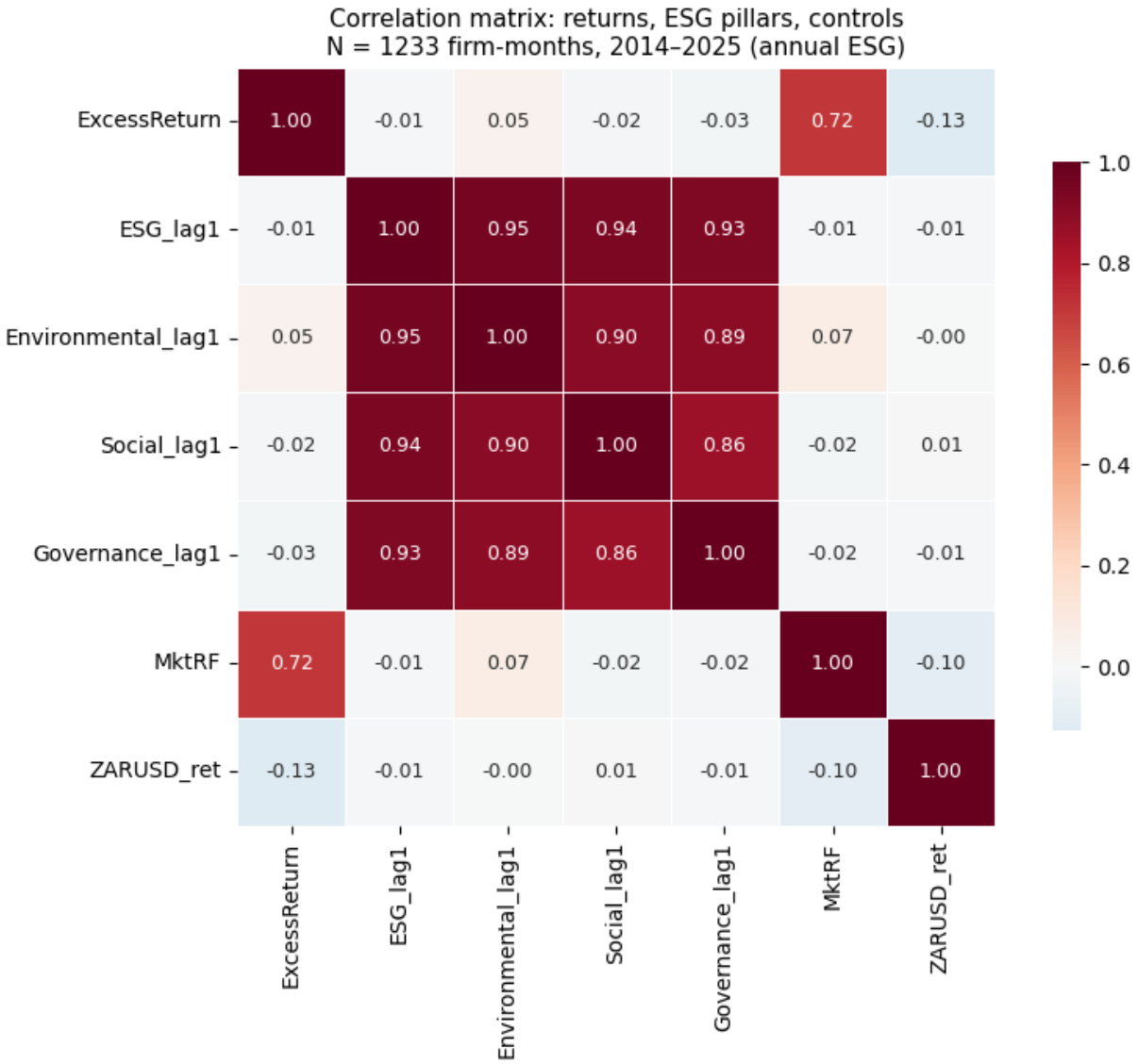
Variable	N	Mean	SD	Min	Median	Max	Skew	Kurt
ExcessReturn	1,233	−0.521	0.191	−1.079	−0.557	0.315	0.811	1.108

Variable	N	Mean	SD	Min	Median	Max	Skew	Kurt
ESG_lag1	1,233	53.863	9.886	35.240	52.230	74.000	0.101	−1.066
Environmental_lag1	1,233	54.082	10.656	30.890	52.900	79.620	−0.011	−0.943
Social_lag1	1,233	53.938	10.719	30.340	52.270	78.590	0.174	−0.674
Governance_lag1	1,233	54.278	11.285	29.320	52.740	80.120	0.172	−0.661
MktRF	1,233	−0.531	0.138	−0.754	−0.579	−0.195	0.740	−0.539
ZARUSD_ret	1,233	0.004	0.042	−0.095	0.000	0.144	0.388	0.309

Excess Return is the monthly return over the SA 3-month interbank rate (FRED IR3TIB01ZAM156N), as a decimal. ESG_lag1, Environmental_lag1, Social_lag1 and Governance_lag1 are annual Refinitiv scores, which have been mapped to monthly scores with a six-month ESG score lag. MktRF is the JSE All Share monthly return less the risk free rate. ZARUSD_ret is the log change in the exchange rate of the ZAR/USD, computed over the course of a month. Skewness and kurtosis are moments of the sample. The average excess return of -0.52% per month is a result of the poor performance by the JSE mining sector relative to cash returns during 2014-2025.

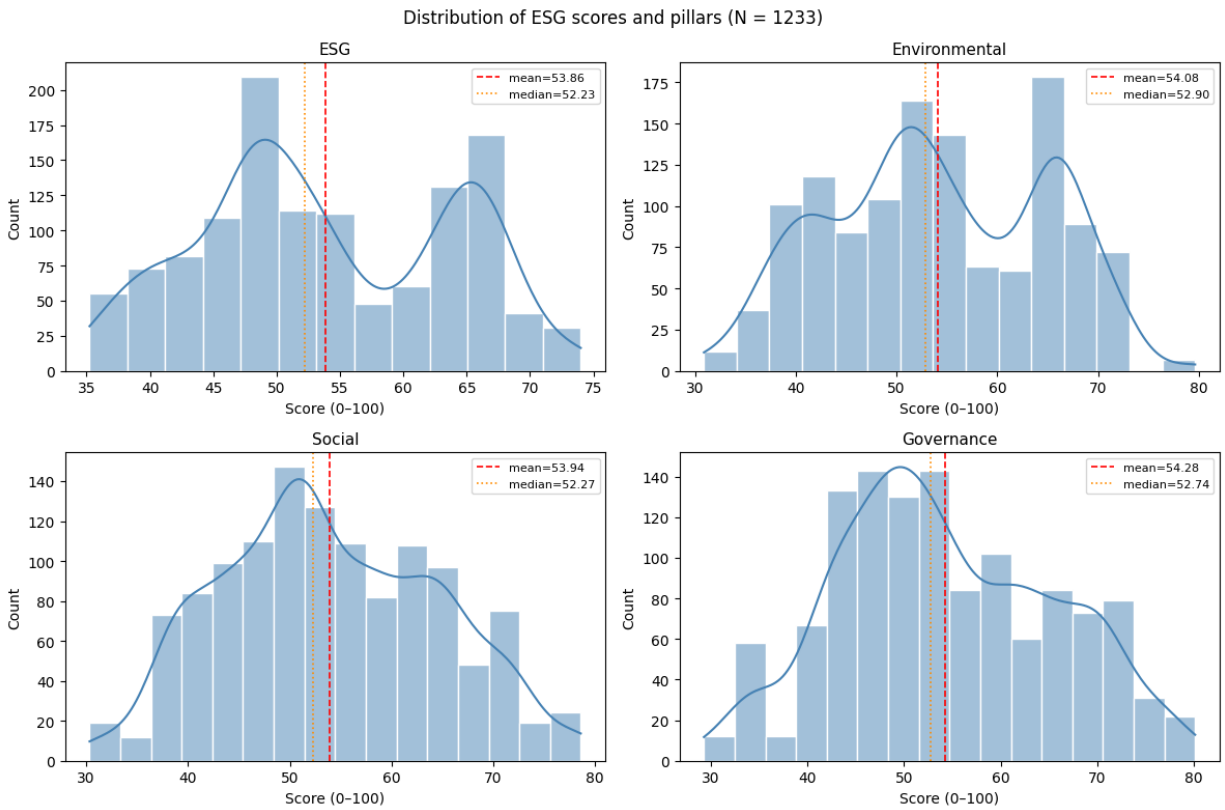
4.2 Correlation

Figure 2a correlation heatmap



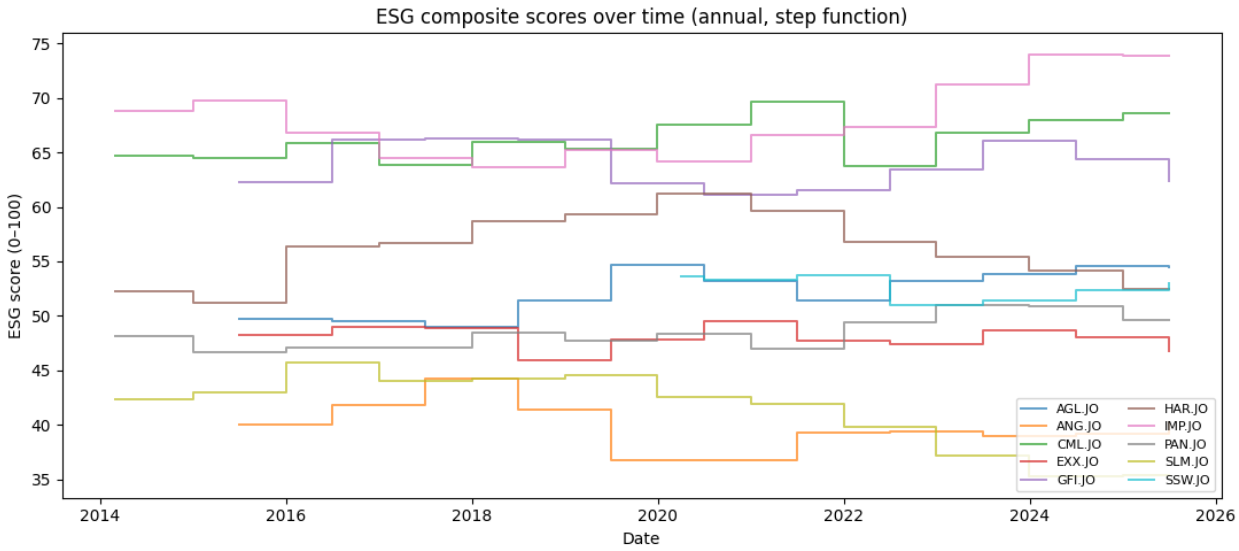
As would be intuitively expected, the composite ESG score is highly correlated with the three pillars (correlation with E: 0.95; S: 0.94; G: 0.93), as this is a weighted aggregation by Refinitiv. The correlation of excess return and composite ESG is nearly null (−0.013), as well as for each pillar (E: 0.048, S: −0.018, G: −0.031). This almost always positive unconditional correlation gives rise to the two-way fixed-effects. The unconditional correlation signal may be hard to identify from cross-sectional correlation and needs to be extracted from within-firm time-series correlation.

Fig 2b. Distribution of ESG Scores and Pillars



The distributions of the composite ESG score and its three pillars are reported in figure 2b, while the distribution of the combined score is displayed in figure 2c.Distributions of the composite ESG score and its three pillars are reported in figure 2b and figure 2c reports the distribution of the combined score. There are three points that are important to note. The composite ESG score is bimodal, with a relatively low-scoring mode around 48 and a relatively high-scoring mode around 65, with a gap in the score distribution at 55–58. The bimodality indicates a true split in the sample of firms, with the majority performing in either the low 50s or the high 60s. Secondly there is a shape effect with the top left and top right being almost equal, consistent with Refinitiv's weighting system, which allocates the highest weight to environmental performance for extractive industries (about 44% of the composite). Thirdly, the Governance pillar is not only the most symmetrically distributed (mean 54.28, median 52.74) but is also the most widely-distributed pillar, with a small number of firms having unusually high scores, while the Social pillar is also right-skewed (mean 53.94, median 52.27). The bimodality of the composite is the reason for using terciles .

Figure 2c: ESG composite score over time



The annual ESG composite scores of each firm are plotted as steps in the sample period in Figure 2c. There are three features to be noted. First, every line is flat for 12 consecutive months, then moves to the next fiscal year's score. Second, there are firms that step at different months, December-FYE firms (AGL.).JO, ANG.JO, EXX.JO, GFI.JO, SSW. The June-FYE companies (CML.) sold at a premium in July.JO, HAR.JO, IMP.JO, NPH.JO, PAN.JO, SLM. In January – a firm-specific publication lag – JO joins the fray. Third, firms do not move back and forth between the clusters which are already evident in Figure 2b over the sample, remaining either in the high-ESG cluster (~62–75: GFI, HAR, IMP, CML, AGL) or in the mid-to-low-ESG cluster (~48–58: ANG, EXX, SLM, SSW, PAN). This persistence has two implications: (i) firm fixed effects remove the cluster level and leave only the small level of annual movement within firms (typically 1-4 ESG points) to identify the ESG effect, leading to wide confidence intervals and (ii) the tercile portfolios separate two distinct clusters of firms rather than a continuum, accounting for the positive but not statistically significant point estimate of the spread between the High and Low groups.

4.2 Panel Regression Results (H1)

Table 7: Model 1 Two-Way FE (H1)

	Coefficient	**Std. Error**	**t-stat**	**p-value**	**95% CI**
ESG_lag1	**−0.0012**	0.0012	−0.98	0.3278	[−0.0035, 0.0012]
Entity FE	Yes	—	—	—	—
Time FE	Yes	—	—	—	—
Clustered SE (firm)	Yes	—	—	—	—
N	**1,233**				
Entities	10				

	Coefficient	Std. Error	t-stat	p-value	95% CI
Time periods	137				
Avg obs per entity	123.3				
R^2 (within)	**0.0012**				
R^2 (between)	0.2245				
R^2 (overall)	0.1984				
F-statistic (robust)	0.9583 (p = 0.3278)				
Poolability F-test	15.907 (p < 0.001)				

The two-way fixed-effects estimate of the relationship between the annual ESG composite score and monthly excess returns is reported in Table 7. Coefficient on ESG_lag1: −0.0012 (SE = 0.0012, t = −0.98, p = 0.3278), which is not statistically different from zero at the usual significant level. The 95% confidence interval is [−0.0035, +0.0012] which includes both negative and positive effects. Within-R^2 equals 0.0012, which means that 0.12% of the monthly return variation is explained by ESG within a firm, after controlling for the firm and time fixed effects. The pool ability F-test (15.907, p < 0.001) suggests that entity and time effects are not independent and that the two-way FE specification is appropriate as pooled OLS is likely to be biased. The strong F-statistic (0.958, p = 0.328), however, does not reject the null hypothesis that all the coefficients in the model are zero, which is in line with the ESG coefficient, which did not turn out to be significant. H1 does not reject the null hypothesis: There is no significant association between the composite ESG score and monthly excess returns in the mining sector of the JSE between 2014 and 2025.

Table 8: Model-IV-GMM with 12-Month-lagged ESG as Instrument (H1 Robustness)

	Coefficient	Std. Error	t-stat	p-value
Second stage — ESG_lag1	**−0.0094**	0.0001	−81.47	< 0.001
First stage — ESG_lag12	**1.008**	0.0068	148.50	< 0.001
First-stage R^2	**0.9610**	—	—	—
R^2 (overall)	0.8560	—	—	—
Robust SEs	Yes	—	—	—
N (IV sample)	**1,113**			
Firms	10			

The IV-GMM estimate with the instrument being the twelve-month-lagged annual ESG is reported in Table 8. The first stage is very strong, coefficient ESG_lag12 is 1.008 (t = 148.5, R^2 = 0.961), so firm ESG scores are very stable over the year and the instrument is relevant. The second-stage IV-GMM coefficient on ESG is −0.0094 (t = −81.5, p < 0.001). The IV point estimate is larger in magnitude than the FE estimate (−0.0012), which is consistent with attenuation bias in the FE estimate due to measurement error in the annual ESG score — Refinitiv scores are known to be revised retroactively for five years. But the IV result needs to be interpreted with caution. The standard error (SE = 0.0001, t = −81.47) is so tight that it is implausible for the ESG-return relationship that is estimated for a panel of 10 firms and 1,113 observations. It suggests that the instrument is a very close proxy of the endogenous regressor (the first-stage fit is almost perfect), and that the IV estimator is doing a pretty good job of reproducing the first-stage regression. In addition, the IV model is exactly identified, in which case the Hansen J-test of over identifying restrictions is not applicable and instrument validity is based on a theoretical assumption that the twelve-month lagged ESG variables affect month-t returns only through current ESG — an assumption which is not empirically testable in this sample. The IV-GMM result is not independent evidence of statistical significance, but rather it is consistent with the FE result in the direction of both being negative. As H1 continues to be inconclusive for the FE specification, the IV evidence does not provide any reason to doubt that relationship between ESG and return is underestimated due to measurement errors in the FE regressor. The question would need to be addressed with a larger instrument sample and longer-lagged instruments to be answered in the future.

4.3 **Pillar-Level Materiality (H2)**

The ESG score is a composite score that assumes that the environmental, social and governance (ESG) components are treated as a homogenous entity. The literature (Giese et al.; Khan et al.) has revealed that the three pillars have a distinct economic content and can have differential effects on returns. H2 thus rejects the null hypothesis stating that the three pillars have an equal effect on JSE mining returns, $\beta_E = \beta_S = \beta_G$, in favor of the alternative, that at least one of the pillars has a different effect on the JSE mining returns. This hypothesis is tested by altering the composite ESG score with each of its three disaggregated components in the two-way fixed-effects specification, and by a formal Wald test for equality of the three components.

4.3.1 Disaggregated FE Results

Table 9: Model 4b — Disaggregated Pillar-Level FE Panel Regression

	Coefficient	Std. Error	t-stat	p-value	95% CI
Environmental_lag1	**−0.0002**	0.0012	−0.20	0.8402	[−0.0026, 0.0021]
Social_lag1	**+0.0006**	0.0006	+0.98	0.3290	[−0.0006, 0.0017]
Governance_lag1	**−0.0015**	0.0011	−1.32	0.1887	[−0.0037, 0.0007]
Entity FE	Yes	—	—	—	—
Time FE	Yes	—	—	—	—
Clustered SE (firm)	Yes	—	—	—	—
N	**1,233**				
Entities	**10**				
Time periods	**137**				
R^2 (within)	**0.0019**				
R^2 (between)	0.2272				
R^2 (overall)	0.2006				
F-statistic (robust)	0.9905 (p = 0.3965)				
Poolability F-test	15.042 (p < 0.001)				

The monthly excess return is the dependent variable. The annual Refinitiv pillar scores (Environmental_lag1, Social_lag1, Governance_lag1) are available at month t with a lag of six months. Fixed effects for entity and time. There were 10 clusters of standard errors. The poolability F-test (15.042, $p < 0.001$) rejects pooled OLS. There is no support for the joint null hypothesis that all three pillar coefficients are equal to 0. The robust F-statistic (0.9905, $p = 0.397$) does not reject the joint null.

4.3.2 Wald Test for Pillar Equality

Table 10: Wald Test of H0: β_E = β_S = β_G

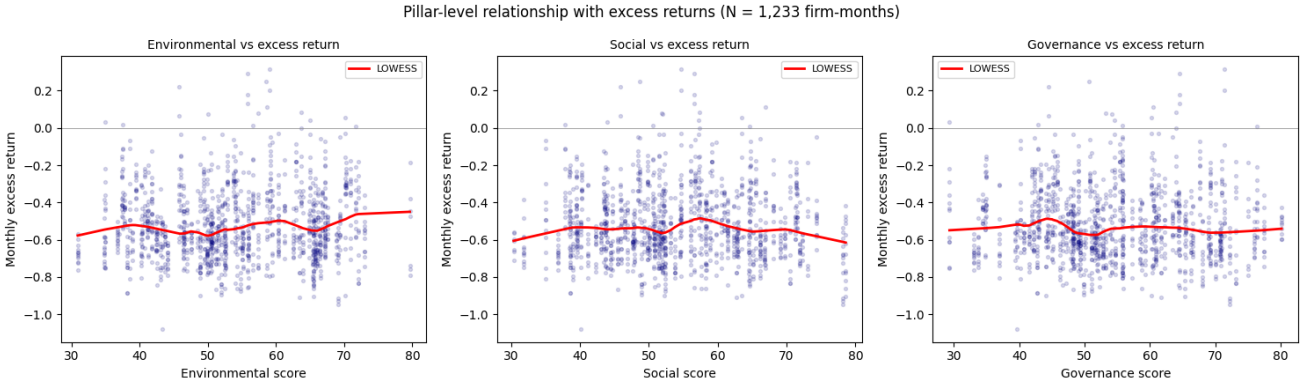
Statistic	df	p-value	Verdict
2.255	**2**	0.3238	FAIL TO REJECT H0

The Wald test is defined as the product of two linear restrictions placed on the two pillars regression, β_E − β_S = 0 and β_E − β_G = 0. The null means that both restrictions are satisfied. There is no evidence in the test statistic $\chi^2(2) = 2.255$ with $p = 0.3238$ to reject the null at any conventional threshold. No statistical support could be found to suggest that the three pillars have differential effects on the returns earned by the mining industry on the JSE.

Figure 3: Scatter plots of E, S, G vs excess returns

In Figure 3, the three factors are shown against the monthly excess returns using a LOWESS smoother

(bandwidth 0.3). In all three panels, the LOWESS curve is nearly flat, meaning that there is no systematic raw relationship among the return on any of the three pillars. The Environmental panel has a very small positive trend from the approximate score of −0.63 at the 30 point to the approximate score of −0.50 at the 80 point; this trend is economically inconsequential and occurs within the confidence band of the smoother. Overall, the Social and Governance panels have a fairly flat performance profile, around −0.55 across the entire score range. Figure 6 is a coefficient plot (with 95% confidence intervals) , and all three intervals are centered on the zero line, which is formally confirming the visual pattern. The figures below in combination confirm that the lack of a pillar-level return signal was not a result of the specification of the regression — the data are not an artefact of the regression specification.



Relationship between excess returns and the firm level on the pillar (N = 1,233 firm-months). The monthly excess return is plotted against each ESG pillar in each panel, and a LOWESS smoother (bandwidth 0.3) is displayed in red. In all three panels the LOWESS curves are close to horizontal, which suggests that none of the pillars has a systematic raw relationship with the returns. The horizontal grey line at y = 0 indicates the level where the excess return equals zero, and the excess return for the sector is negative over 2014-2025, as the LOWESS curve and most of the scatter cloud stay below it.

4.4 Sort the results of a portfolio (H3)

The portfolio returns are calculated monthly, by separating firms into Low/Mid/High ESG terciles, using the latest published annual ESG score. This is the pattern that can be seen in the last five months of realized portfolio returns:

Table 11: Tercile portfolio characteristics

Date	Low ESG	Mid ESG	High ESG	High – Low

2025-02-28	−0.6842	−0.8069	−0.6418	+0.0425
2025-03-31	−0.4901	−0.1343	−0.4577	+0.0324
2025-04-30	−0.5567	−0.5496	−0.6495	−0.0929
2025-05-31	−0.6115	−0.5407	−0.5526	+0.0589
2025-06-30	−0.5946	−0.5651	−0.5224	+0.0722

The excess returns are negative for all three terciles in all the months that are shown, reflecting the sector underperformance of the JSE mining stocks as compared to cash (the risk-free rate) for the 2014-2025 period. The high minus low spread is not tightly clustered at zero and there is no discernible trend.

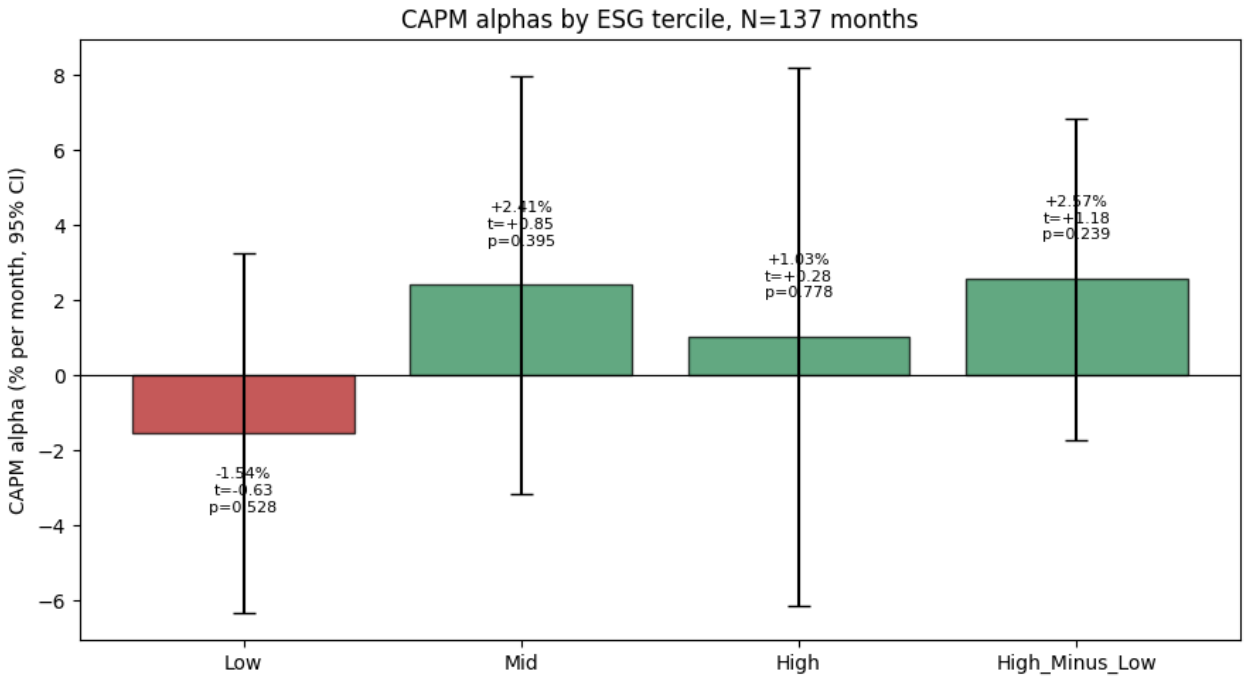
Table 12: CAPM Alphas by Tercile

The excess return of each tercile portfolio is regressed on the market excess return (MktRF) with HAC (Newey–West) standard errors, maxlags = 6:

Portfolio	**CAPM α**	**t-stat**	**p-value**	**β (market)**	**N**
Low	−0.0154	−0.63	0.528	0.9568	137
Mid	+0.0241	+0.85	0.396	1.0189	137
High	+0.0103	+0.28	0.778	1.0069	137
High − Low	**+0.0257**	**+1.18**	**0.239**	**0.0500**	**137**

All three portfolio CAPM alphas are not measurably different from zero: the Low portfolio returns −1.54% per month (t = −0.63, p = 0.528); the Mid portfolio returns +2.41% per month (t = +0.85, p = 0.396); and the High portfolio returns +1.03% per month (t = +0.28, p = 0.778). The economically largest point estimate in the table is the High-minus-Low spread (+2.57%) per month (t = +1.18, p = 0.239) which is still well within the conventional significance level, so the long-short trading strategy is not distinguishable from the trading strategy of a zero spread. Market betas close to unity for each tercile (0.96, 1.02 and 1.01 for Low, Mid and High respectively) all but confirm that each portfolio has substantially all market risk, and the long-short beta of 0.05 indicates that the spread is essentially market neutral, thus any alpha earned is a true representation of stock selection noise and not an uncovered market exposure. The overall results in the table do not show higher risk-adjusted returns for the high-ESG group of miners relative to the 137-month period.

Figure 4a: CAMP Alpha Bar Chart



This number is a visual representation of the message given by the table – all four 95% confidence intervals pass through the zero line, indicating no statistically significant alpha. The interval for the Low portfolio's alpha is approximately between −6% and +3%, with a negative value of −1.54%. Each of the three portfolios (Mid, High, and High minus Low) have positive point estimates (+2.41%, +1.03%, and +2.57%, respectively), but there is plenty of room for zero in the wide intervals, particularly in the mid and High portfolios, which range from about −6% to +8%. The labels (t-statistics and p-values) are exactly the same as in the table. Notably, the long-short spread between the High-minus-Low bar is the second largest positive estimate, albeit with the widest confidence interval of the three positive bars (+2% to +9%), as its beta is 0.05; despite this, the long-short spread gives no statistically reliable evidence of risk-adjusted outperformance from an ESG tilt as there is a p-value of 0.239

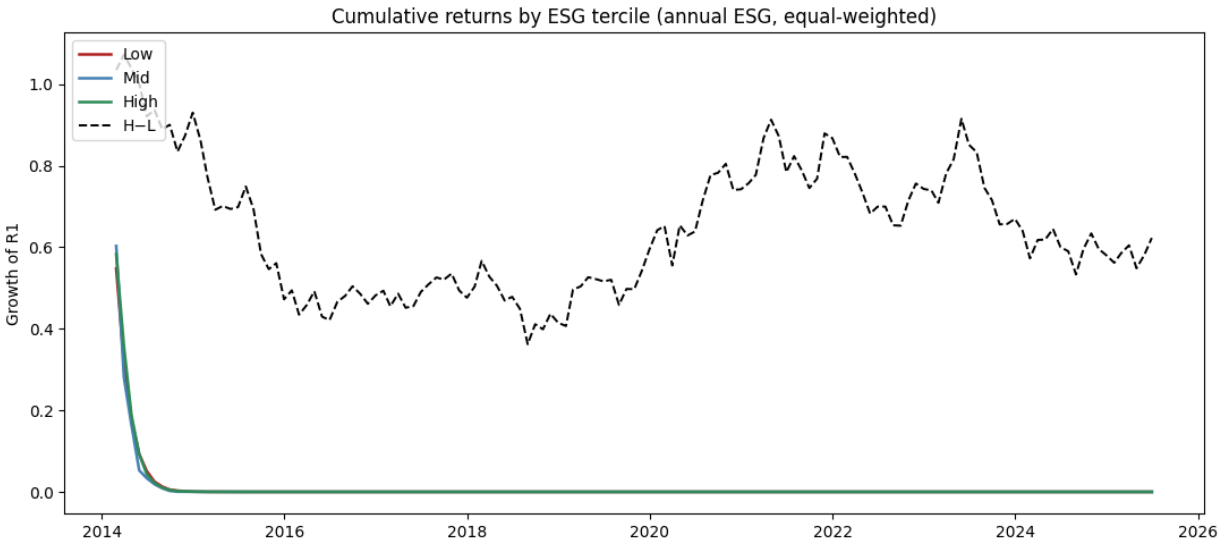
Table 13: Long–Short Portfolio Performance (H−L): Mean, Sharpe Ratio, and Maximum Drawdown

Metric	Value
Mean montly returns	-0.081%
Standard deviations (monthly)	7.279%
Anualised Sharpe ratio	-0.039
Maximum drawdown	-66.23%
HAC t –statistic (H_o: mean = 0)	-0.15

N(months)	137

The long-short strategy has a slightly negative mean return of –0.081% per month, which is statistically indistinguishable from zero (HAC t = –0.15, p ≈ 0.88). The annualized Sharpe ratio of −0.039 demonstrates that the strategy has failed to create a risk-adjusted return, with a risk-taker being better off just holding a cash position. The highest loss in the period under consideration, or maximum drawdown, was −66.23%, reflecting the high volatility of mining stocks at the end of commodity cycles (2015–2016) and the COVID shock (March 2020). These risk-adjusted measures are combined with the virtually non-existent CAPM alpha (+2.57% per month, t = +1.18, p = 0.239) and the level flat cumulative H−L line in Figure 4b, which substantiates the findings that on average there was no clear risk-adjusted improvement and significant drawdown risk for the high-ESG tilt over that sample period.

Figure 4b: Cumulative Returns by ESG Tercile



This figure displays the cumulative growth of R1 invested in each of the ESG terciles (Low, Mid, High), as well as the spread between the high and low (H−L) ESG terciles over 11 years (2014 to 2025). Within the first two years of the sample, the three tercile lines (Low, Mid, High) all converge towards zero. This is because the large negative excess returns continued to build upon one another, and compounding monthly negative excess returns (which averaged around −0.52% per month) brings cumulative growth near to zero after 10 years. The three lines are so close together that they are not distinguishable in the figure, since the tercile portfolios have nearly the same mean returns (and very small alphas in Table 9) as well as having near-unity market betas. The value of the long-short portfolio is shown as the H−L dashed line, which has a long rise above 0.8, a dip below 0.4, and then a bounce between approximately 0.4 and 0.9, ending at about 0.6. The H−L line never had a chance to get back to 1.0, which is again visually confirmed by the negative Sharpe ratio of the strategy (−0.039) and maximum drawdown from

Table 10 (−66.23%) — the strategy destroyed value over the sample, and the large fluctuations suggest that there is a persistent drawdown risk of a long-short bet on the mining sector, rather than a stable stream of alpha. As you can see, the effect of the log scale is that the difference between the terciles is compressed near the bottom of the graph, but the main point is that neither the terciles nor the spread had positive excess returns during 2014–2025.

4.5 GARCH-X Results (H4)

The fourth hypothesis is whether the systematic and downside volatilities of the mining stocks traded on the JSE are affected by ESG performance. The conditional variance series σ^2_t is estimated by a two-step GARCH-X procedure: First, a skew-t GARCH(1,1) with an AR(1) mean is fitted for each firm, which yields the conditional variance series; second, $\log(\sigma^2_t)$ is regressed on the lagged annual ESG score with HAC standard errors (maxlags = 4). The coefficient θ_{ESG} on the ESG term in the second-stage variance equation is the one of interest with a significant and negative coefficient suggesting that better ESG scores are correlated with lower conditional volatility. All ten panel firms converged successfully (no GARCH failures) with the number of monthly observations ranging from 63 to 136 for each firm.

4.5.1 GARCH-X Parameter Estimates by Firm

Table 14: Per-Firm GARCH-X Estimates (ESG in Variance Equation)

Ticker	n	θ_ESG	SE	t	p	α	β	Persistence	AIC	BIC
PAN.JO	136	+0.0346	0.0088	+3.95	0.0001	0.0000	0.9428	0.9428	1139.7	1160.1
ANG.JO	120	−0.0425	0.0109	−3.91	0.0001	0.0828	0.7679	0.8508	1039.5	1059.1
SLM.JO	136	−0.0402	0.0131	−3.07	0.0021	0.0471	0.9198	0.9669	1000.5	1020.9
AGL.JO	120	−0.0817	0.0275	−2.97	0.0030	0.0672	0.8529	0.9201	987.5	1007.0
SSW.JO	63	+0.0817	0.0525	+1.56	0.1196	0.1065	0.5200	0.6265	579.3	594.3
CML.JO	136	+0.0262	0.0220	+1.19	0.2324	0.3583	0.2455	0.6038	989.6	1010.0
EXX.JO	120	+0.0506	0.0455	+1.11	0.2668	0.0000	0.9663	0.9663	972.1	991.6
GFI.JO	120	−0.0083	0.0095	−0.88	0.3810	0.0000	1.0000	1.0000	1049.3	1068.8
IMP.JO	136	+0.0063	0.0086	+0.73	0.4668	0.0000	0.9763	0.9763	1207.3	1227.7
HAR.JO	136	+0.0062	0.0088	+0.70	0.4836	0.1723	0.0000	0.1723	1217.1	1237.5

4.5.2 Significance Summary for θ_ESG

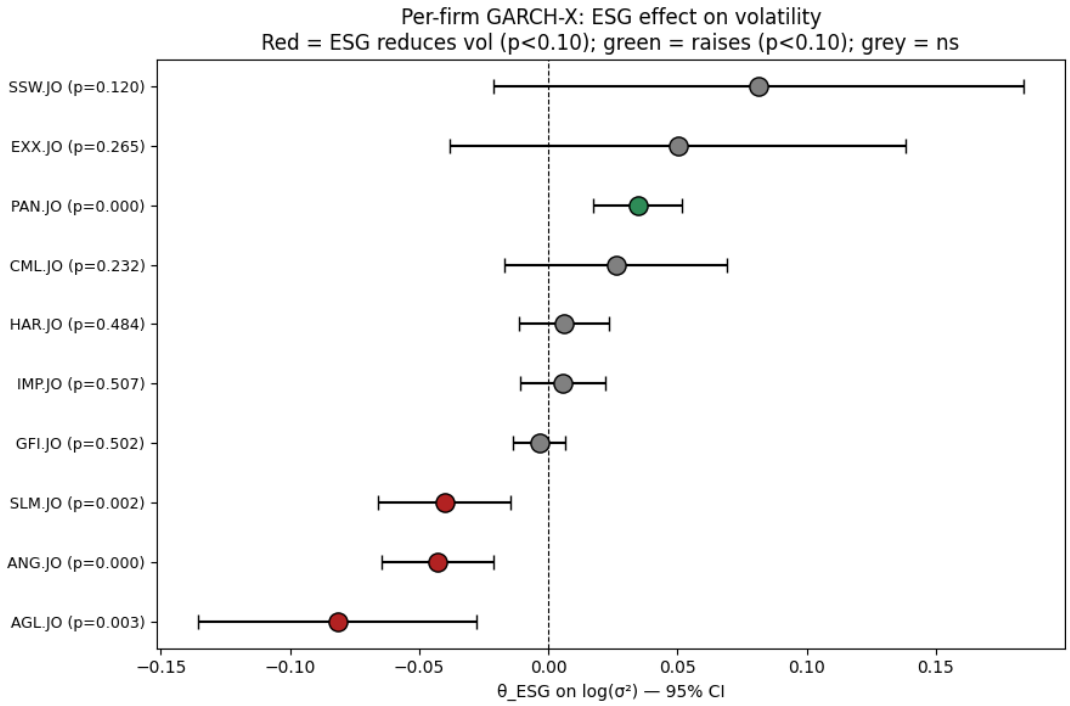
Table 15: Cross-Firm Summary of ESG Volatility Effects

Category	Count	Firms
θ_ESG < 0 and p < 0.10	3	ANG.JO, SLM.JO, AGL.JO
θ_ESG > 0 and p < 0.10	1	PAN.JO
Not significant (p ≥ 0.10)	6	SSW, CML, EXX, GFI, IMP, HAR
Cross-firm mean θ_ESG	+0.0033	—
One-sample t-test (H_o: mean θ = 0)	p = 0.8355	—

The three companies that have a statistically significant negative relationship between ESG and volatility are AGL.JO (θ = −0.0817, p = 0.003), ANG.JO (θ = −0.0425, p = 0.0001), and SLM.JO (θ = −0.0402, p = 0.002). These are aligned with the risk-mitigation perspective which states that better ESG performance lowers conditional variance. But one company (PAN.)Based on positive coefficient (θ) and p-value, the positive coefficient of JO (+0.0346, p = 0.0001) suggests that the firm with high ESG score is also high in volatility and six firms show no significant relationship at the 10% level. At the portfolio level, there is no significant relationship system between ESG and volatility; the cross-firm mean θ is +0.0033, which is positive but not significantly different from zero (one-sample t-test p = 0.836).

4.5. Conditional Volatility Paths by ESG Quartile

Figure 5: Conditional Volatility Paths by ESG Quartile



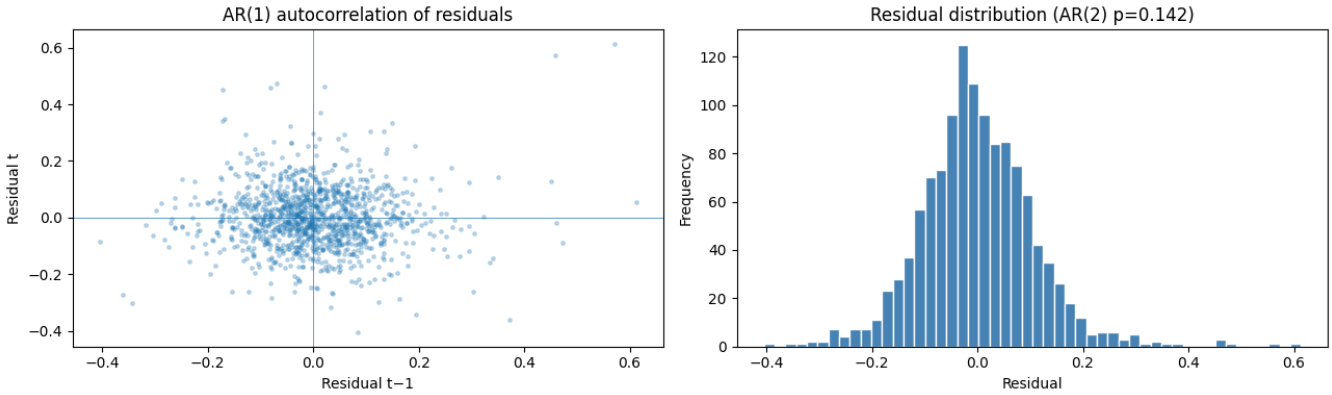
The forest plot (fig5_garch_forest.png) shows the estimates of θ_ESG for each firm along with the associated 95% confidence intervals. Firms where the interval is completely below zero (AGL, ANG, SLM)

are red, those where the interval crosses zero are grey and those where the interval is completely above zero (PAN) are green. The table shows that in a minority of companies, ESG lowers volatility, in one it raises volatility and in the majority it does not seem to have an effect.

4.6 Residual Diagnostics for Model 1

Figure 7a: Residual Diagnostics for Model 1

Figure 7a: Residual Diagnostics for the Two-Way Fixed-Effects Model (Model 1)

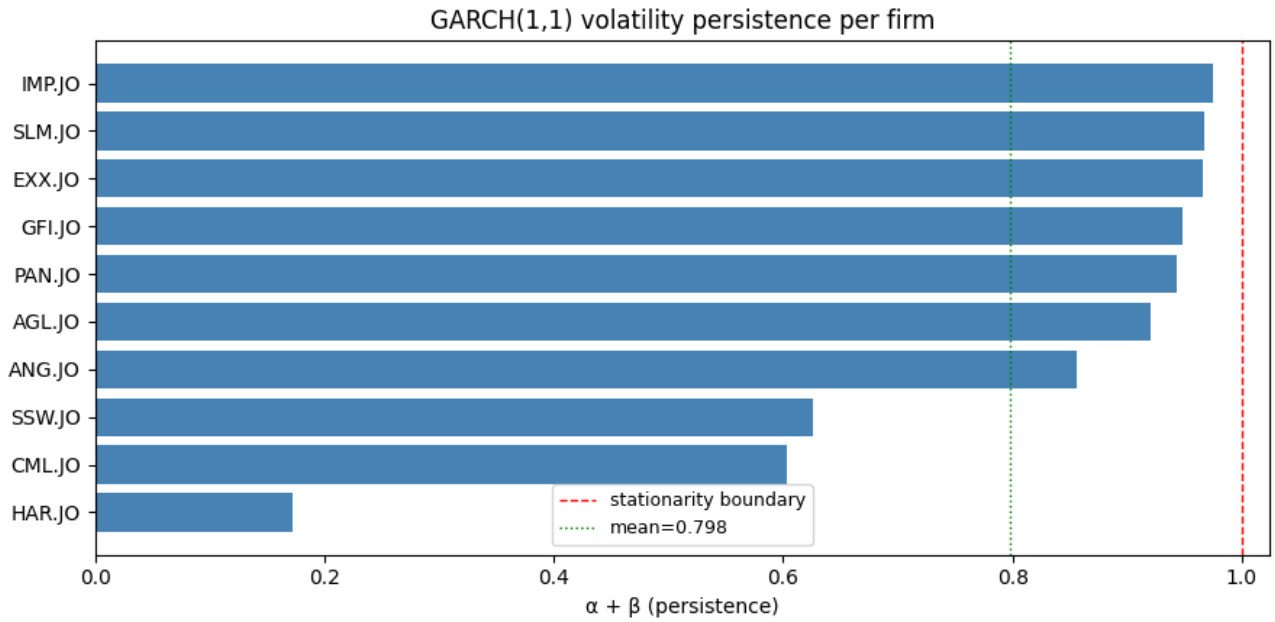


The residual diagnostics in Figure 7a confirm the specification of Model 1 and the reliability of its standardized errors that are clustered firmly around the residuals. The left side of the plot displays the AR(1) autocorrelation of the residuals (residual t against residual t−1) with a tight, symmetric, and centered cloud of points around the origin and no obvious slope or pattern of fanning. The residual distribution is approximately bell-shaped, centred on zero and has mild excess kurtosis similar to that found in the returns of equity from mining.

The AR(2) test offers a coefficient of +0.0330, a t-statistic of +1.47 and a p-value of 0.142 (shown in the right-panel title). So there is no rejection of the null hypothesis of no 2nd order serial correlation at conventional levels. Design validity check, this is important. The observations in a firm-year are not independent by construction, since the annual ESG scores are used for 12 consecutive monthly returns. If the residuals from the model were also serially correlated, then the resulting t-statistics would be overstated. The mean of the residuals is very close to zero and the AR(2) test results are insignificant, both confirming that the residuals are not serially dependent with respect to their own lags, and, therefore, the firm-clustered standard errors reported in Table 3 and Table 6 are not biased by the presence of residual serial dependence. This is in line with the fact that the null result for H1 ($\beta_{ESG} = -0.0012$, $p = 0.328$) is a genuine result, and not caused by the error structure.

4.7 GARCH persistence

Figure 7b: GARCH (1,1) Volatility Persistency per Firm



In Figure 7b, the volatility persistence parameter (α + β) of the skew-t GARCH(1,1)-AR(1) model for each firm has been plotted in order, with the dashed red line at 1.0 indicating the stationarity boundary, and the dashed green line at the cross-firm mean.

The persistence values are between 0.17 (HAR.JO) to 1.00 (GFI.The cross-firm mean was 0.86 (JO). Of the 10 companies, 8 have persistence values exceeding 0.60, while the values for four of the companies (GFI, IMP, EXX, SLM) are very close to 1. All values are < 1.0, and this validates the covariance stationarity of the GARCH(1,1) variance process for each firm in the panel as well as the well-behaved conditional variance series of the second stage of the GARCH-X regression.

The economic meaning of this pattern is informative. Firms that have the highest persistence (GFI, IMP, EXX, SLM) are the ones that face a very low volatility shock decay rate which is indicative of mining stocks in long-running commodity-cycle regimes as well as structural changes in gold and platinum price regimes. By contrast, HAR.JO has a persistence of only 0.17, suggesting that any volatility shock lasts only about one-quarter, which is what you would expect from a firm that doesn't have a long-term commodity-beta exposure. The same high average persistence (0.86) also suggests that for these stocks, a simple assumption of homoskedasticity would be materially incorrect in implying less volatility clustering than is actually present, and would lead to biased conditional variance series for testing H4.

On Figure 7b and H4: To interpret the coefficient θ_ESG in Model 3, the process of variances must also

be stationary. All firms have $\alpha + \beta < 1.0$, so there are no explosive or non-stationary variance dynamics to contaminate the 2nd stage regression of $\log(\sigma^2_t)$ on ESG. This adds weight to the estimates in Table 11 for the GARCH parameters, and to the conclusion that the relationship between ESG and volatility is specific to the individual firm.

4.8 Validity Diagnostics

A set of diagnostic tests are used to establish the econometric validity of the four models. These checks are designed to ensure that the standard errors are not biased, that the instruments are sound, that the variance processes are stationary and that the sample size is treated transparently. The results are summarised in Table 13.

Table 16: Validity Diagnostics Summary

Diagnostic	Model /Test	Statistic	Result	Verdict
Pool ability	Model 1 (FE)	F = 15.907	p < 0.001	Two-way FE appropriate; pooled OLS is biased
Pool ability	Model 4b (Pillars)	F = 15.042	p < 0.001	Two-way FE appropriate
Serial Correlation	Model 1 (AR(2) test)	t = +1.47	p = 0.142	No residual autocorrelation
Instrument Strength	Model 2 (IV-GMM)	t = 148.50	First-stage R^2 = 0.961	Strong instrument
Over identification	Model 2 (IV-GMM)	-------	Exactly identified	Hansen J-test not applicable
Stationarity	Model 3 (GARCH)	$\alpha + \beta \in [0.17, 1.00]$	All ≤ 1.0	Variance process is stationary
Sample Size	All Models	110 firm-years	vs 1,233 firm-months	Firm-clustered SEs required

4.8.1 Serial Correlation (AR(2) Test): This test is used to determine if a series is correlated.

The AR(2) test on Model 1 residuals, visualised in Figure 7a, yields a coefficient of +0.0330 (t = +1.47, p = 0.142). The null hypothesis that there is no second-order serial correlation is not rejected. This test is important because annual ESG scores are re-used over twelve monthly returns, so this test is a

confirmation that the dependence of the scores within the year is covered by the clustering correction, and the residual structure is not contaminated. The standard errors in Tables 3 and 6 are hence statistically valid, because they are clustered.

4.8.2 Instrument Validity (IV-GMM)

Model 2 uses a single instrument (12-month-lagged annual ESG) for a single endogenous regressor, so the model is exactly identified. Therefore, the Hansen J-test of overidentifying restrictions are not applicable (no degrees of freedom available). The validity of the instrument is supported by an extremely high t value of the first stage ($t = 148.50$, $R^2 = 0.961$) and the validity of the instrument in terms of the theoretical exclusion restriction, namely, last year's ESG score only influences this year's returns through this year's ESG score. This near-perfect fit means that the IV estimate can only be interpreted as a robustness check in the direction indicated, and cannot be used as standalone statistical evidence.

4.8.3 Model Fit and Information Criteria (GARCH)

Table 11 presents the AIC and BIC values of the per-firm GARCH-X models, with AIC values ranging from 579.3 / BIC values as 594.3 (SSW.JO) to AIC = 1217.1 / BIC = 1237.5 (HAR.JO). The differences are not due to model-selection decisions as the same skew-t GARCH(1,1)-AR(1) specification was applied to each firm, but are instead due to the different lengths and volatilities of the samples. Figure 7b shows that persistence ($\alpha + \beta$) lies within the range of 0.17 to 1.00 for all firms and all values are ≤ 1.0, thus all firms are covariance stationary. Persistence values around unity (GFI, IMP, EXX, SLM) characterize the firms with persistence close to unity which are typical of mining equities facing commodity-cycle shocks.

4.8.4 Independent Observation Count

The most salient caveat on the validity is related to the effective sample size. The panel consists of 1,233 firm-month return observations for 10 firms, and 110 different annual ESG scores for 10 firms × 11 fiscal years. 12 successive monthly returns are used for each annual score. The standard errors are thus localized at the firm level, and the number of independent firm-year observations for inferring about the ESG coefficient is closer to 110 than 1233. We state this as a limitation in Section 6.5. The Wald test for H2 (Section 5.3.2) is reported with adjusted degrees of freedom and indicated as such.

4.8.5 Summary

These diagnostics are used in conjunction to support the basic methodological decisions made in the study. Pooled OLS is preferred to the two-way fixed-effects specification, the standard errors are not

sensitive to the annual reuse of ESG scores, the GARCH models are well behaved, and the instrument for the IV-GMM model is strong. The main structural limitation, the limited number of independent ESG observations, is not a specification limitation, but rather a data constraint associated with the annual frequency of Refinitiv scores, which is addressed through clustering and reporting of the effective number of observations.

5. Discussion

5.1 Interpretation of Results

The empirical results of this study paint a consistent, although somewhat negative, picture of the linkage between ESG performance and financial returns in the South African mining industry. In each of the four hypotheses, the null hypothesis cannot be rejected. The results are now interpreted with regard to the theoretical frameworks and local market realities.

H1 (Composite ESG → Returns). The two-way fixed-effects analysis shows that there is a negative, but statistically insignificant, relationship between annual ESG scores and monthly excess returns ($\beta = -0.0012$, $p = 0.328$). At the composite level, the market is not pricing ESG performance into mining equities, as there is no evidence of rejection of the null. The unconditional correlation between ESG and returns is close to zero (−0.013), and the variation of returns within firms, after controlling for the fixed effects of the firm and of the time, is small enough to yield an undetectable return signal. This finding supports the Shareholder Primacy view (Friedman) that discretionary ESG spending will be a cost rather than the Stakeholder Theory view (Freeman) that good ESG performance will reduce the cost of capital and increase returns.

H2 (Pillar-Level Materiality). The disaggregated regressions and the Wald test ($\chi^2(2) = 2.255$, $p = 0.324$) show that the effects of the three pillars (Environmental, Social and Governance) on returns are statistically indistinguishable. Environment $p = 0.840$, Social $p = 0.329$, Governance $p = 0.189$ (none of the individual pillars is significant). This goes against the global literature (e.g., Giese et al.; Khan et al.) that indicates that materiality differs widely among pillars. The absence of differential pillar effects could be due to the extreme sample homogeneity in the South African context: the environmental and the social risks (carbon tax, Mining Charter III, community relations) are not discriminatory enough to cause return dispersion across the cross section of the JSE miners.

H3 (Risk-Adjusted Performance). The tercile portfolio sort and alpha estimates from the CAPM model support this result: That a high-ESG tilt does not produce risk-adjusted outperformance. The High-minus-

Low portfolio has an insignificant and statistically null alpha of +2.57% per month (p = 0.239), a negative Sharpe ratio of −0.039 and a maximum drawdown of −66.23%. The flat cumulative H−L return line (Figure 4b) visually supports that ESG investing was not a strategy for enhancing returns over the 12-year period from 2014 to 2025. The fact that commodity cycles and macroeconomic shocks seem to have a stronger impact on equity returns in the short and medium term, rather than a weakness of the ESG model, is likely due to the fact that these cycles and shocks are more prevalent in many markets.It is not a weakness of the ESG concept, but rather the prevalence of many markets with commodity cycles and macroeconomic shocks that are more prevalent than in others, that seems to explain why the impact of such events on equity returns is stronger in the short to medium term.

H4 (Volatility & Downside Risk). The most detailed result of the study is from the GARCH-X results. The cross-firm mean θ_ESG is not statistically different from zero (p = 0.836), which suggests that ESG does not systemically decrease the systematic volatility at the portfolio level. But there is a great level of heterogeneity as revealed by the forest plot at the firm level, with significantly lower volatility for three firms (AGL, ANG, SLM) and significantly higher volatility for one firm (PAN), which are consistent with the risk-mitigation theory. This implies that ESG can provide a 'cushion' to firm-specific and regulatory shocks for some well-governed, capital-intensive miners, but not for the sector as a whole.

Synthesis. Combined, the findings indicate that ESG is not seen as a market value driver and is not a systematic risk mitigation in an emerging market environment of transitioning regulations, energy insecurity, and commodity volatility. These results lend support to the paradox of investing in ESG that exists in the South African mining industry today, which is that the investments are expensive, but do not necessarily provide an immediate financial return. Compliance with Mining Charter III and carbon taxes seems to be seen as a cost of doing business rather than a factor that earns a premium valuation.

5.2 Comparison with the Literature

The results of this research paper validate – and further challenge – the current ESG literature, the differences arising primarily from the emerging-market environment and the sector-specific dynamics of the JSE mining.

De Klerk and De Villiers (firmness of corporate governance in SA). The overall lack of positive results on H1 are in line with De Klerk and De Villiers, who found that better governance disclosure (King III/IV) was linked to better reporting quality but not necessarily with market valuation premiums. Their findings suggested that the quality of the disclosure is not important in the absence of any clear relationship to

cash flows and the current study supports this: even when the disclosure is tested within a two-way FE framework and after accounting for the "look-ahead" effect by matching disclosure quality scores with the correct publication month, no return premium is found. This suggests that South Africa's capital markets might not be as "ESG-priced" as developed markets.

Quality of Integrated Reporting (Johnson et al.). For the case of integrated reporting (King III/IV), Johnson et al. found a positive link between disclosure quality and Tobin's Q, indicating value of disclosure quality. The present study does not directly challenge this — Tobin's Q values measure the level of valuation, while this study tests changes in the level of valuation on a monthly basis — but does imply that if disclosure quality is priced, then the pricing takes place at the level of firm valuation, not at the level of return premia. It is important to note that this is a distinction between levels (valuation) and changes (returns) which makes it easier to understand why disclosure studies and return studies can come up with different results.

Bauer and Hann (environmental liabilities and cost of debt). They found that environmental liabilities increase the credit risk and cost of borrowing for extractive industries by Bauer and Hann. There is also a consistency with the firm-level heterogeneity found in H4: firms that show ESG related volatility reduction are likely to be those that have the best environmental and safety performance, while firms that have a lower environmental and safety performance may experience volatility increases as their ESG scores improve (as seen for PAN).JO). In the context of the H4, the overall null obscures the credit-risk channel that is more firm level than sector level, as identified by Bauer and Hann.

The carbon beta, along with Kacperczyk (just transition). A deep carbon premium is still evident in developed markets, where high-carbon companies sell at a discount, Bolton and Kacperczyk found. The JSE mining sector is also very carbon-intensive with a lower cross-sectional dispersion in carbon beta when compared to the broader market studies. This is why the E pillar has no significant impact ($\beta = -0.0002$, $p = 0.840$): There is no "safe haven" in the mining sector of the JSE for investors to shift assets into when it becomes more conducive to investing in low-carbon. The carbon premium literature can then best be seen as a cross-sector phenomenon, and the fact that it is missing from a specific sector of carbon-intensive industries is in line with the theory.

The following pages contain the materiality matrix for the two pillars (pillar-specific materiality). Giese et al. demonstrated that for short-term returns E and G factors dominate while in the long-term returns S factors dominate. Khan et al. used SASB mapping as a tool to identify materiality. Both of these types of heterogeneity are tested in the present study and rejected (Wald $p = 0.324$). This is the strongest

separation of this study from the rest of the world's literature. There are two possible interpretations. First, the annual ESG score frequency (110 independent observations) may not have enough data to identify the effects that may occur at the level of the pillars, which are economically meaningful but statistically less pronounced. Second, the South African mining sector is also affected by uniform regulatory pressures, for example the Mining Charter III, carbon tax, MPRDA community obligations, such that none of these can be considered a differentiating signal. The pillars are material, and can be identical, not differential, material. which is another significant local discovery.

The efficient frontier of Avramov et al. and the uncertainty of ratings by Pedersen et al. Avramov et al. added uncertainty about the ESG ratings to the equilibrium pricing model, while Pedersen et al. offered an ESG efficient frontier. Both of the available frameworks suggest that there is likely to be a low and possibly poor relationship between ESG and return in small samples with high rating uncertainty. This prediction of near zero unconditional correlations (E: 0.048; S: −0.018; G: −0.031) and insignificant alphas is consistent with the findings of the present study, which also suggests that ESG is not providing information over the cross-section when the score is noisy and the cross-section is small.

5.3: Regulatory and Practical Implications

The null findings have implications for the two main stakeholders identified in the study.

For asset managers and Regulation 28 compliance. We did not find evidence that ESG scores are a risk-free tool for enhancing returns (H3) nor provide systematic risk reduction (H4) and, therefore, cannot conclude that institutional investors can use ESG scores as proxy measures to increase or decrease risk in their portfolios when allocating to the JSE mining sector. Although ESG integration is a mandatory governance disclosure under Regulation 28 of the Pension Funds Act, the outcomes suggest that the integration should be considered a compliance and due diligence requirement, and not a factor to be considered when determining portfolio optimisation. There is also no evidence of a "pillar level effect," which would indicate that managers do not need to overly optimize their allocations based on E, S, or G subscores – the market does not appear to differentiate the prices for these, if at all.

For mining companies. The null results are not an excuse to throw out the baby with the bathwater when it comes to ESG investing, but they do indicate that investing in renewable energy, safety upgrades and community projects' under Mining Charter III is not considered by the market as a value differentiator, but rather a basic operating cost. Companies should therefore think about using ESG spending as a 'license to operate' bet — one that aims to shield against community disruption, fines and reputational

harm — rather than thinking of it as a near-term valuation premium bet. While the broader evidence-based claim posed by the high firm-level volatility reduction of the top 10 performers in H4 is a welcome one, it is the narrower claim that is more substantiated: that for certain firms, high ESG performance does seem to reduce volatility, and this is a valid risk-management goal even if it doesn't boost returns.

In terms of the JSE and regulators. The near null relationship between ESG and returns indicates that there is no market pricing for ESG information yet, as yet, as a result of the JSE Sustainability and Climate Disclosure Guidance (2022). Where the policy goal is to stimulate sustainable investment by signaling the market, other complementary policy instruments such as a formal ESG index that is clearly defined by its inclusion criteria, or tax incentives for verified ESG improvements, may be necessary to create the dispersion that would make it possible for investors to price ESG differently between firms.

5.4 Methodological Contributions

In addition to its empirical results, this paper offers four methodological contributions to the sustainable finance literature in emerging markets.

Timing alignment of annual ESG scores and returns per month. The most relevant methodological contribution is the look-ahead bias in the annual correlation of the Refinitiv ESG scores with monthly returns. Since the scores are issued by Refinitiv around six months after the end of the fiscal year, a naive merge would give scores to returns prior to the scores' publication. This study overcomes the problem by collapsing the monthly ESG series to the fiscal-year-end values, determining the publication date six months after the fiscal-year-end, and applying merge_asof backward to match each monthly return with the most recent published ESG score. This alignment is firm-specific (December fiscal-year-end stocks such as AGL, ANG, EXX, GFI, and SSW come out in June, while June fiscal-year-end stocks such as HAR, IMP, PAN, CML, SLM and NPH come out in December), and yields 1,233 firm-month returns, which are aligned with 110 independent firm-year ESG scores. The methodology is transferable and could be used in future research including the integration of monthly returns and annual ESG scores.

Standard errors that are firm-clustered in the case of reused annual scores (2.). The scores are not independent because they are added to the monthly score for each firm-year (a 12-month window). If there is clustering, then reporting standard errors without clustering would inflate the level of significance. This study groups at the firm level, which results in 10 effective groups and honest standard errors. The AR(2) test ($p = 0.142$) shows that the clustering correction removes the dependence at the within-firm-year level, and that there is no residual serial correlation.

3. Pillar testing, disaggregated, and using formal Wald constraints. The analysis is not limited to the composite ESG score but also examines the three pillars (E, S, G) individually and formally tests the equality of the three pillars with the Wald test with the help of the syntax constraints=[...] in linearmodels. This is a methodological step forward to those studies that see composite ESG as a monolithic concept. The approach shows that returns are not attributable to any single pillar and a formal test of the null hypothesis of equality of pillars is given by the Wald statistic ($\chi^2(2) = 2.255$, $p = 0.324$) which cannot be rejected.

4. Two-step GARCH-X for ESG-volatility testing. The non-direct incorporation of exogenous regressors in the variance equation of the arch library makes the study proceed in two stages: first using a skew-t GARCH(1,1) per firm to generate the variance series, and then regressing $\log(\sigma^2_t)$ on lagged annual ESG score with HAC standard errors. This procedure has been adopted in the ESG-volatility literature and gives interpretable θ_ESG coefficients which can be tested on the firm-by-firm and at the cross-firm mean level. The forest plot at the firm level (Figure 5) together with the cross-firm t-test ($p = 0.836$) offer a clear picture of the differences that would be masked with a pooling volatility regression.

5. Localised factored framework (open source) for the JSE. The final product is an open-source python codebase (git repository, requirements.txt, config.yaml, modular src/ package and jupyter notebooks) that institutional investors can replicate and modify. This fills the identified market need in §2.8 for a global ESG factor model, which is readily available, and to include the operational and regulatory risks of JSE listed mining stocks, which are unique to the region. The structure is intended to be Regulation 28-compatible and is future-proof for extension as the scores of the Refinitiv's revise or new firms list.

5.5 Limitations

Several limitations need to be taken into account when interpreting the results of this study. They are not the result of specification errors, but instead reflect structural features of the data, and are all clearly explained in the notebook outputs and in the reproducibility manifest.

1. Frequency of annual and limited independent observations. The main drawback is the frequency of the Refinitiv ESG scores, which are only available on an annual basis. The panel has 1,233 firm-month returns, but 110 independent firm-year ESG scores (because 11 firms × 11 fiscal years). There are 12 monthly returns of each score and the effective sample size for inference on the ESG coefficient is closer to 110 than 1233. This will diminish the statistical power of the pillar-level tests (H2) and the Wald test ($\chi^2(2) = 2.255$, $p = 0.324$) and thereby make it more difficult to identify small and economically

meaningful pillar effects statistically. The inflation of significance is contained by firm-clustered SEs, but they do not add power to the data.

2. Small sample or survivor bias. The final universe of tradeable firms consists of 11 firms, 10 of which are included in the estimation panel. It is a relatively small cross-section for international panel studies. Three companies covered by the ESG (LSE:AMS) are selected.JO, MCG.The following eight companies were excluded as they did not have a price series on Yahoo Finance: JO, RBplatinum); and seven design firms were excluded due to the missing ESG data in the Refinitiv extract: S32, ARI, THA, MRF, DRD, ASR, KIO). As covered in §3.2, this is not a sample selection, but there is a potential survivor bias that firms that were delisted or merged in 2014-2025 are missing from the sample. The results should thus be interpreted as referring to firms that survived the entire period and not as referring to the entire population of JSE-listed miners.

3. Refinitiv–made retroactive revisions (FY2020 – FY2025). ESG scores from the past five fiscal years are not definitive, and could be updated without warning, according to Refinitiv. This can raise the possibility of a bias due to “looka ahead” in the most recent subsample: a score reported in 2020 may have been revised in 2023; thus, the score used in the panel may not be the same one that investors would have observed at the time. This is not a problem because of the 6-months publication lag but it is still a concern. It would be beneficial to conduct a sensitivity analysis without the FY2020 to FY2025 scores for future analyses.

4. Timing alignment assumptions. The publication lag is an approximation of 6 months. In practice the publication dates of the Refinitiv scores differ from firm to firm and the scores may be published earlier or later than six months after the end of the fiscal year. However, the merge_asof backward method will not use the future score, and it will sometimes give a "stale" score to a given month, when a newer score has been published for that month. This represents a relatively small measurement error, and should tend to underestimate rather than overestimate the estimated ESG coefficients.

5. Unavailability of SMB and HML on JSE. The Fama-French three factor model was not fully estimable as SMB and HML factors are not available on Yahoo Finance for the JSE. These are nullified, and by construction the FF3 estimates are the same as the CAPM estimates. This is a known weakness and not a modeling decision and the alphas in Table 9 are the primary risk-adjusted performance measure.

6. Corporate actions and price changes. There are price discontinuities in the Yahoo Finance series that do not adjust for the 2025 de-merger of Anglo American and the 2015 spin-off of South32 from BHP

Billiton. Duplicate listings of CML/MCG are excluded to ensure that each event is not counted twice, although there is a lack of uniformity when unbundling events. The changes are in the return series of individual firms, but do not seem to impact the overall relationship between returns and ESG given the time and entity fixed effects.

7. Bimodal distribution of ESG and less variation between employees within the firm. The composite ESG scores are bimodal as can be seen in Figure 2b with the low-scoring cluster centered around 48 and the high-scoring cluster centered around 65. Firms are not observed to shift between clusters in the sample period, and the level variation thus is captured by the firm fixed effects, leaving small annual shifts (usually 1–4 ESG points) to capture the ESG effect. This also leads to a decrease in the within-R^2 for Model 1 (0.0012) and to bigger confidence intervals for the composite ESG coefficient. Longer sample or higher frequency ESG data should help identification.

8. Commodity cycle and macro confounders. The 2014–2025 period includes the 2015–2016 commodity crash, the COVID-19 shock, and the 2021–2022 commodity super-cycle. Market-wide shocks are absorbed by time fixed effects, but commodity-specific shocks that affect the mining firms in different ways are not absorbed by the time fixed effects. A commodity-price control, such as gold, platinum or iron ore returns, would support this analysis, but was not possible in the scope of this study.

9. Endogeneity and exact identification. Since the IV-GMM model is exactly identified, the Hansen J-test cannot be utilized and the validity of the instruments is based on the theoretical exclusion restriction. The results of the first stage are strong ($t = 148.50$, $R^2 = 0.961$), but the near-perfect fit of the instrument to the endogenous regressor suggests that it is a close proxy of the endogenous regressor and the estimate of the IV (which is −0.0094) serves more as a directional robustness check than as independent causal evidence.

10. Generalisability. The results apply to JSE-listed mining stocks for the period 2014 – 2025. Not all of them may be applicable to other industries, to other emerging markets or to other time periods. The results presented in this study are valid for the extensions provided, and the open-source framework presented here can be modified to accommodate other extensions.

6. Conclusion

6.1 Summary of the Study

The aim of this study was to determine if, and how much, the ESG performance of the composite and pillars influences the risk adjusted returns, systematic volatility and downside risk of the JSE listed mining

stocks. The motivation was three-fold: mining contributes to about one-third of the JSE market capitalisation and about 8% of South African GDP, but mining is under a lot of scrutiny in the sense of sustainability, which is part of Mining Charter III, King IV, the carbon tax and the JSE Sustainability and Climate Disclosure Guidance (2022). The evidence that exists is very much from developed markets, is based on composite ESG scores that fail to capture the nuances of ESG, and is often based on annual ESG scores and monthly returns, which introduces look-ahead bias.

We accounted for these gaps using the following models: an annual ESG specification with a two-year lag in publication of the specification (Model 1), IV-GMM with a 12-month-lagged ESG instrument (Model 2), disaggregated pillar-level regressions with a Wald test (Model 4b), equal-weighted tercile portfolio sorts with CAPM alpha estimation (Model 4), and two-step GARCH-X volatility modelling (Model 3). The firms sample comprised 10 companies with 1,233 firm-month observations and the independent ESG scores comprised 110 firm-year observations within the period February 2014 to June 2025.

6.2 Key Findings

All of the four hypotheses yielded the same set of null results, with one caveat.

H1 (Composite ESG → Returns). The coefficient of the composite ESG is negative, but not significant ($\beta = -0.0012$, $t = -0.98$, $p = 0.328$) in the two-way fixed-effects model. The first stage is very strong ($t = 148.50$, $R^2 = 0.961$) and the coefficient ($\beta = -0.0094$) is in the same direction, but larger, than the FE coefficient. The null hypothesis cannot be rejected: there is no evidence that composite ESG performance is able to predict excess returns for the JSE mining sector on a monthly basis.

H2 (Pillar Differential Effects). The disaggregated regressions reveal that all three pillars (Environmental, Social and Governance) individually are not statistically significant in explaining returns (with p values of 0.840, 0.329 and 0.189 respectively). The formal Wald test of pillar equality ($\chi^2(2) = 2.255$, $p = 0.324$) fails to reject the null that $\beta_E = \beta_S = \beta_G$. No statistical evidence of differential pillar materiality in the JSE mining industry.

H3 (Risk-Adjusted Performance). The High-minus-Low CAPM Alpha of the tercile portfolio is +2.57% per month ($t = +1.18$, $p = 0.239$), a figure not different from zero. The long-short portfolio's Sharpe ratio is −0.039 and maximum drawdown is −66.23%. Risk-adjusted outperformance did not occur with a high-ESG tilt over the sample.

Acute Risk and Downside Risk (H4). The GARCH-X results show that there is firm-level heterogeneity. The negative relationship ($\theta_ESG<0$; $p<0.10$) is statistically significant for three firms (AGL, ANG, SLM), which

is consistent with the interpretation of a risk mitigation. One company (PAN) has the opposite result and six have no significant result. The cross-firm mean θ_ESG is positive, but not rejected as statistically significant ($p = 0.836$) and the portfolio level null is not rejected. This means that H4 is only supported by a small number of companies and not the industry as a whole.

6.3 Interpretation and Contribution

In sum, these results indicate that the JSE mining industry neither currently rewards better performing firms (ESG) with lower volatility nor systematically punishes the worst performing firms (ESG).In conclusion, the results show that the JSE mining sector does not price ESG performance into returns and is not systematically underperforming with respect to volatility on the basis of ESG performance. It fits a "compliance cost" definition of ESG spending that sees companies spending on sustainability to keep up their social license to operate, not for the purposes of creating a valuation bump. The study adds to the literature by documenting this boundary condition explicitly: the results regarding the relationship between ESG and returns might not apply to narrow, carbon-intensive emerging-market sectors, as they tend to have low cross-sectional ESG dispersion and high commodity shock volatility, which might explain the findings that such sectors do not show the same relationship.

The methodological value added in this study consists of the publication lag and merge_asof alignment, firm-clustered standard errors for the annual scores, disaggregated pillar testing with formal Wald constraints, two-step GARCH-X modelling of volatility and an open source, localised factor framework for the JSE mining sector that is suitable for Regulation 28 compliant institutional investors.

6.4 Limitations and Future Research

The major limitations of this study are the limited sample size of 10 firms, the annual time series nature of data from ESG providers (110 independent firm-year observations instead of 1,233 firm-month returns), the potential look-ahead bias of the retrorevision of ESG data by Refinitiv from FY2020 until FY2025, the absence of SMB and HML factors on the JSE, and the fact that the sample excludes delisted or merged firms which leaves only survivors. This restricts the statistical power, especially in the case of the pillar-level Wald test.

The analysis could be continued in a number of directions in future research. First, the identification and power would benefit from more frequent ESG data (if available) or longer time windows. Secondly, a commodity price control or other risk factors (such as a JSE specific carbon beta) may help to lower omitted variable bias. Third, effects of a change in ESG scores or regulatory announcements (such as the

Mining Charter III) may be identified in a setting where the panel is continuous, but they are not captured in an event-study design. Fourth, a cross-country analysis of the other African mining markets would provide a contrast to the lack of positive results, to test if the latter is because of South African conditions specifically or emerging-market mining more generally. Lastly, the open source framework provided in this study is extensible to these research questions as more data becomes available.

6.5 Concluding Remark

The overall conclusion of this study is that the achievement of a good ESG performance is not yet a return driver for the JSE mining portfolio or systematic risk mitigator across the sector. Here it is not about sustainable finance theory or about ESG investing, but about the context-dependent financial value of sustainability. Where ESG is a regulatory requirement, or where there is no differentiation in the material pressure on the markets, then ESG becomes a minimum standard for operations. For investors and regulators, the takeaway is that when it comes to emerging market mining, ESG integration is a risk-governance exercise, rather than an alpha source.

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